

Network Traffic Analysis Report

Analyst: Nandini Adhikari **Date:** 9/26/2026 **Capture File:** 2026-09-11-traffic-analysis-exercise.pcap

Tool Used: Wireshark

1. Executive Summary

This report documents the analysis of a packet capture involving authentication and directory-service activity between two hosts on a Windows Active Directory network. The investigation traced a user account's activity from initial login through several follow-on service calls, evaluating each for signs of malicious behavior. While one line of investigation (directory replication access) was ruled out as benign, a pattern of repeated account/group-membership queries was identified that is consistent with either legitimate administrative activity or post-compromise reconnaissance. The capture alone does not provide enough context to distinguish between the two.

2. Scope and Methodology

Analysis began with an overview-first approach rather than sequential packet review:

1. **Statistics -> Protocol Hierarchy:** established which protocols were present and in what volume
 2. **Statistics -> Conversations / Endpoints:** identified the primary hosts involved
 3. **Targeted filtering:** investigated specific protocols of interest (Kerberos, DCERPC/DRSUAPI, SMB2, SAMR) based on findings from step 1
 4. **Packet-level inspection:** examined individual requests/responses only after filters had narrowed the scope
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3. Hosts Involved

IP Address	Role
10.9.11.135	Client host: initiates authentication and subsequent service calls
10.9.11.2	Domain controller: receives authentication requests, hosts DRSUAPI/SAMR services

4. Protocol Overview

Protocol Hierarchy analysis showed a mix consistent with a Windows domain environment, including:

- Kerberos (authentication)
- LDAP (directory queries)
- SMB / SMB2 (file/session access)
- DCE/RPC, including DRSUAPI and SAMR interfaces
- NetBIOS (legacy name resolution)
- TLS (77.3% of total bytes was bulk of traffic volume, not part of this investigation's scope)

The presence of DRSUAPI and SAMR traffic were interfaces not typically used by standard client activity and was the basis for deeper investigation.

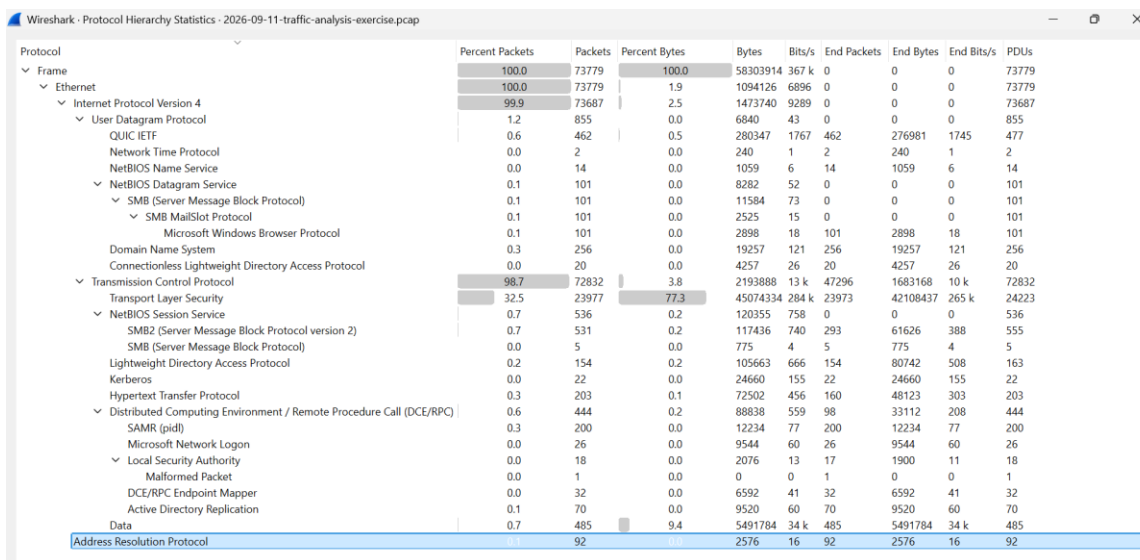


Figure 0: Protocol Hierarchy Statistics for the full capture

5. Findings

5.1 Authentication (Kerberos)

- **Account observed:** mcdowell
- **Realm:** OVERHANDS.ORG
- Multiple AS-REQ packets were observed in rapid succession (four packets within ~0.1 seconds).

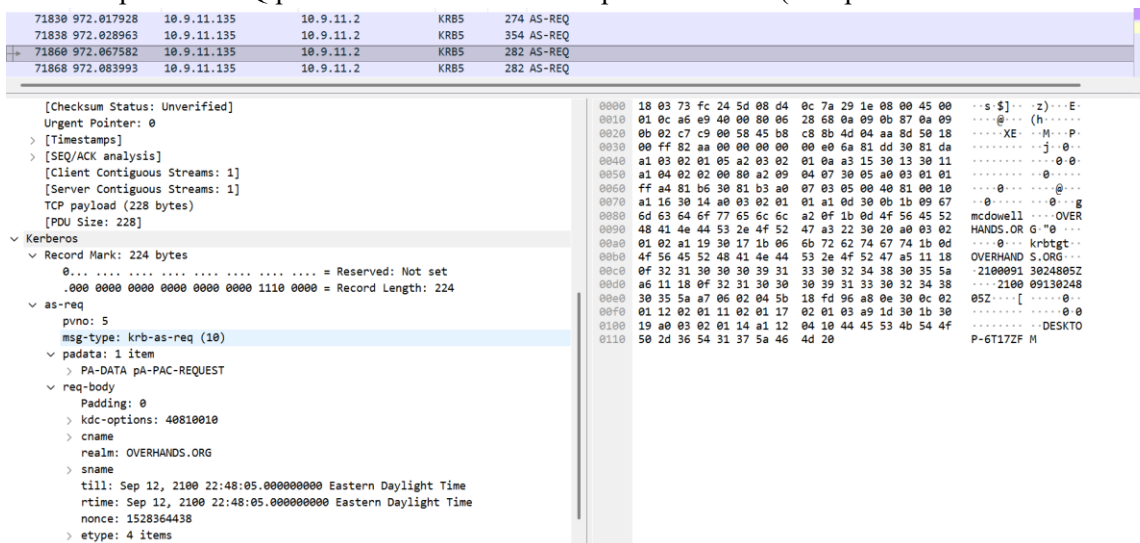


Figure 1: Kerberos AS-REQ packets showing account mcdowell, realm OVERHANDS.ORG

- Initial assessment suspected this pattern indicated brute-force login attempts. Further inspection of the response codes showed the error KRB5KDC_ERR_PREAUTH_REQUIRED; this is

standard Kerberos pre-authentication behavior, not a failure. No instances of KRB5KDC_ERR_PREAUTH_FAILED (which would indicate an incorrect password) were found.

- **Conclusion:** Authentication for mcdowell completed successfully via normal protocol behavior. Initial suspicion of brute-forcing was not supported by the evidence.

5.2 SMB2 Session Establishment

- Following successful Kerberos authentication, 10.9.11.135 established an SMB2 session to 10.9.11.2.
- Session Setup Response returned a success status.
- The session connected to the IPC\$ administrative share and opened a handle to the samr named pipe.

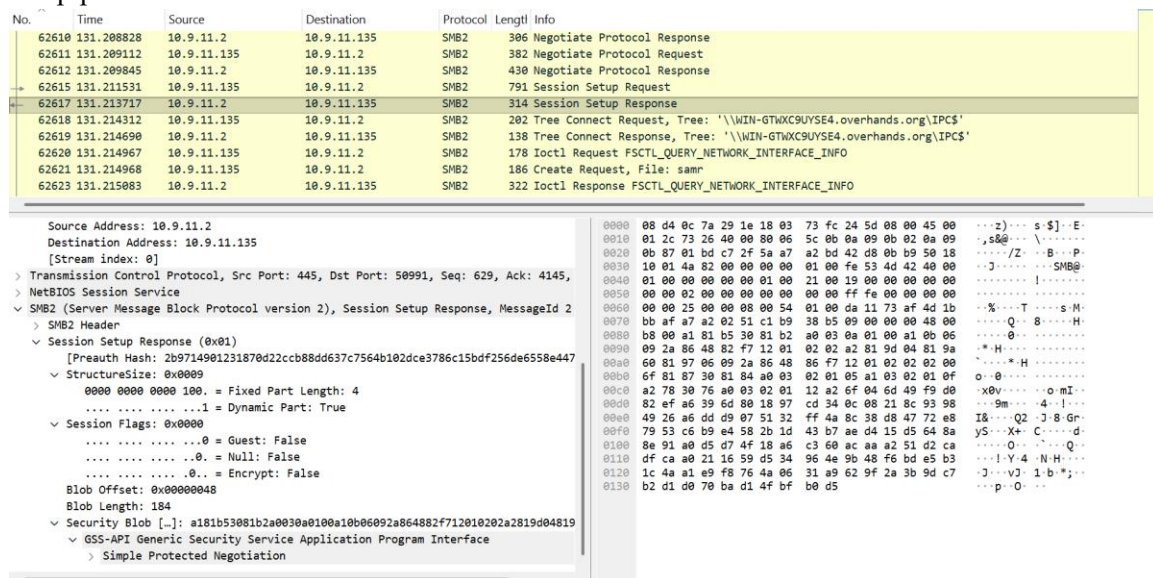


Figure 3: SMB2 Session Setup Response, Tree Connect to IPC\$, and Create Request for samr

- **Conclusion:** Session establishment was successful and proceeded directly to SAMR access, which prompted further investigation (see 5.3).

5.3 Directory Replication Service (DRSUAPI)

- DCE/RPC Endpoint Mapper queries confirmed a lookup for the DRSUAPI service, followed by an active DRSUAPI session (DsBind -> operation -> DsUnbind).
- The specific operation used was **DsCrackNames**, a standard name-format translation function (e.g., converting a username to a distinguished name).

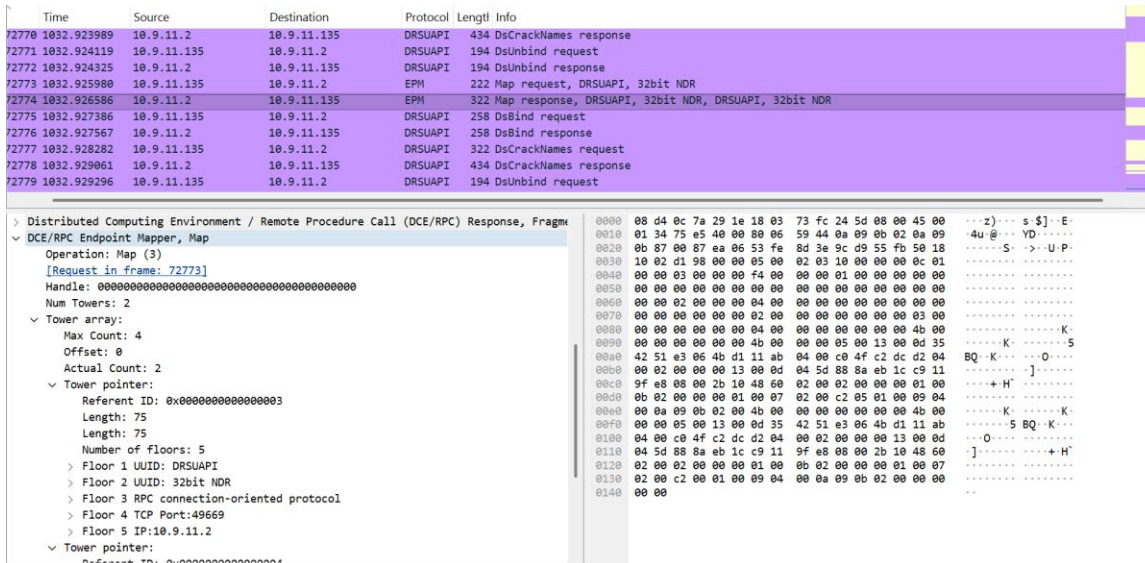


Figure 4: DRSUAPI session showing DsBind -> DsCrackNames -> DsUnbind (no DRSGetNCChanges present)

- The capture was checked specifically for **DRSGetNCChanges**, the operation associated with directory replication/credential extraction (commonly referred to as a “DCSync” attack). This operation was **not present** anywhere in the capture, including within Wireshark’s dedicated “Active Directory Replication” protocol grouping.
- Conclusion:** DRSUAPI activity was benign. No evidence of directory replication abuse or credential dumping.

5.4 Account/Group Enumeration (SAMR)

- After establishing the SMB2/SAMR session, the client issued **six GetAliasMembership requests** over a span of approximately four minutes (timestamps ~578s, ~806s, ~972s).

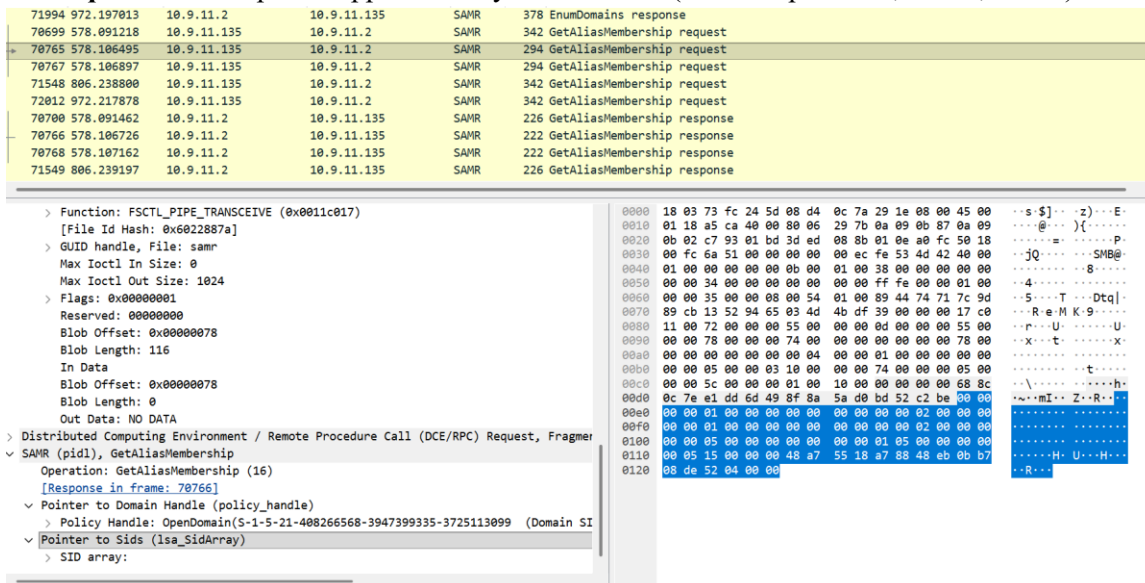


Figure 2: SAMR GetAliasMembership requests/responses (frames 70699-72012)

- GetAliasMembership queries which groups/aliases a given SID (account) belongs to.

- The specific SID(s) queried and the group memberships returned were not conclusively identified during this analysis and would require further inspection of the SID array fields in the request/response packets.
- **Conclusion:** This repeated, time-spread querying pattern is consistent with either:
 - Legitimate administrative tooling (e.g., a helpdesk script or AD management tool checking account permissions), or
 - Reconnaissance activity following account access, aimed at determining what privileges the authenticated account holds.
 - The capture alone cannot distinguish between these two explanations; this would require additional context (e.g., whether mcdowell is an IT/admin account with a legitimate reason to run such queries).

6. Indicators Observed

Type	Value	Notes
Account	mcdowell	Authenticated account of interest
Domain/Realm	OVERHANDS.ORG	
Client IP	10.9.11.135	Initiated all activity described above
Server IP	10.9.11.2	Domain controller
Hostname	WIN-GTWXC9UYSE4.overhands.org	Target of SMB2 session

7. Overall Assessment

The traffic reflects a fully successful authentication chain (Kerberos -> SMB2 -> SAMR) by the account mcdowell against the domain controller. Two initially suspicious leads which were rapid Kerberos requests and DRSUAPI access were both investigated and found to be benign upon closer inspection. The repeated GetAliasMembership calls remain a point of interest that could not be fully resolved from packet data alone and would be the recommended focus of any follow-up investigation (e.g., correlating with host-based logs or confirming the account's intended role).

Recommendation: In a live environment, this activity would warrant a follow-up with the account owner or system logs to confirm whether mcdowell's SAMR queries were authorized administrative activity.

8. Lessons Learned (Analyst Notes)

- An "Error" label in a protocol's Info column does not always indicate failure — KRB5KDC_ERR_PREAUTH_REQUIRED is a normal part of the Kerberos exchange, not evidence of a failed login.
- Two packets appearing close together in time are not necessarily related; conversation context (matching IP pairs, stream index) must be confirmed before drawing conclusions.
- Verifying the specific operation/opnum within a protocol (e.g., DsCrackNames vs. DRSGetNCChanges) is necessary before labeling protocol activity as malicious — the interface being used is not sufficient evidence on its own.
- Overview tools (Protocol Hierarchy, Conversations, Endpoints) are significantly more efficient starting points than sequential packet review for a capture of this size.